07 July 2026 06:53

DS Workflow

1. Business Domain
2. Data collection
3. Data Understand head() tail des info()
4. Data Preprocessing
   - → handle Null
   - → remove duplicate
   - → handle outliers
   - → handle skewed
   - → handle categori
   - → Scaling — Normali, Stand, Robust
5. EDA → checking Relatn ship b/w the vari
6. Feature Engg → remove, add, extract → col
7. Split the data
8. Classificatn if data Imbalanced → Smote tech

   if data balanced → Train data
9. Model creation
10. Build & train the Model — Train data ACC
11. Model Evaluation → Performance — Test data ACC
12. Conclude Overfitting under, Genera
13. hyper parameter → model, CV Evaluation → score
14. Build the Model for hyper parameter

LR →
LogR →
Svm → C, γ
DT → max depth
Rand →
XGB

(14) Build the Model for hyper parameter

(15) Overfitting & Underfitting — Bagging (All Algo)
— Boosting (DT).

Rand → XGB

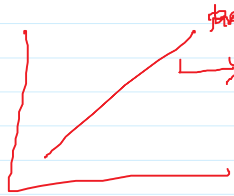
Linear Reg :- Objective : Best fit line

(1) mini Error / Residuals

$\boxed{\text{Residual} = y_A - y_P}$

d_1

RSS ⇒ min value
select as best fit line



best → $\boxed{y = m_1 x_1 + m_2 x_2 + C}$

II Gradient descent tech.

Constant → dataset.

line → Eq ⇒ m, c, $\boxed{x_1, x_2}$

y ⇒

find optimal value of m & c ?
→ based on min error.

m=3.2 c=4.1 optimal m & c

m? c?

Trai / error		
m	c	Error
1.2	1.7	0.2
1.9	2.8	0.5
3.2	4.1	0.01
4.2	4.1	0.75

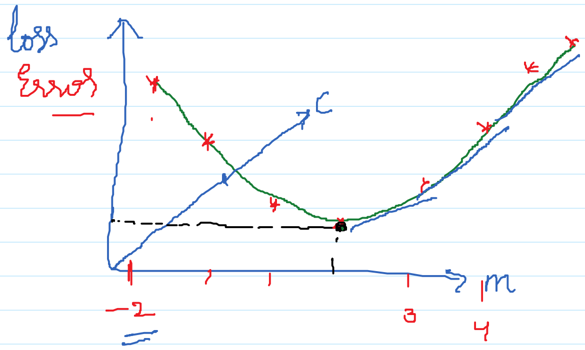
Gradient

loss

Gradient des

Mathemat

partial do



3D

$\tan = \frac{opp}{adj} = \frac{loss}{m}$

$\boxed{\frac{d\,loss}{dm} =}$

m=2 → optimal of m → min Error

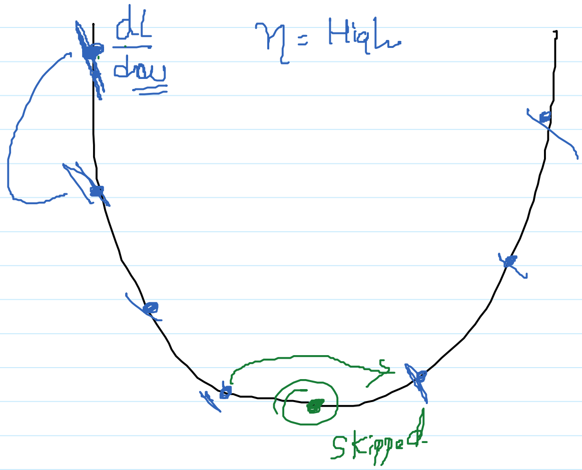
Grad

LR

SVM → SVC

SVR

ANN



$\boxed{\eta = 0.01}$ Small Steps

η → learning

XGB   learning rate = 0.01

$\eta \times [\ ]$

0

① Gradient descent Tech ⇒ Optimization Tech to reduce error

② Regularization ⇒ Prevent overfitting < Train acc ✓ / Test acc ✗

(2) Regularization ⇒ Prevent overfitting < Train data ✓ / Test data ✗

Cost function / loss / Residual = $y_A - \hat{y}$

$\boxed{\text{Cost function = Error}}$ → without Regularization

With Regular

Cost function = Error + $\boxed{\text{Regularization Term}}$

(1) V.V.V.P I 10Q → Exam: Same 5 question → 100%   Penalty

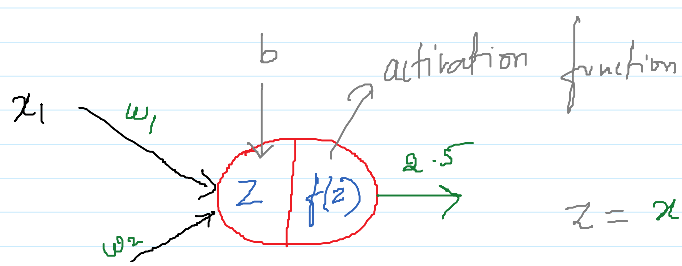
II 10Q → 5 Qp → very very tricky puzzle out indirect } 50%

punish → Mistake

Concept } 100%

Regularization
- L_1 (Lasso) → Remove less important features
- L_2 (Ridge) → instead of removing col. ↓ Priority / importance / weightage to each

$C_1 \to 0.1$ ✓
$C_2 \to 0.8$ ✓
$C_3 \to 0.6$ ✓
$C_4 \to 0.2$ ✓



$z = x_1 w_1 + x_2 w_2 + c$

$z = x_1 w_1 + x_2 w_2 + c$

$z = 2.5$

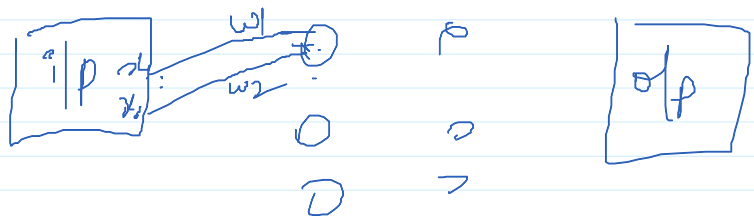
$y = m_1 x_1 + \cancel{m_2 x_2 + c}$

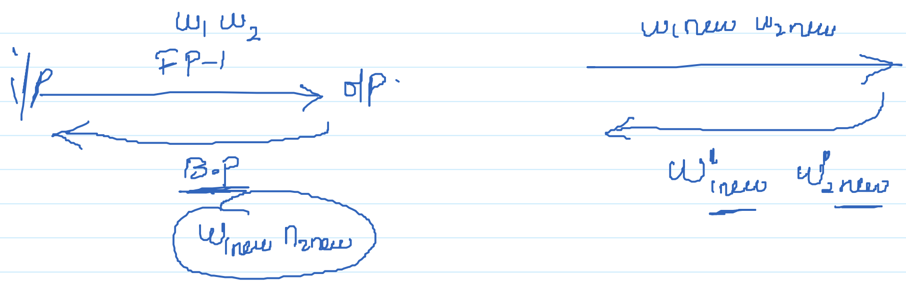
Relu or Tanh ← hidden layer neuron? ✓ ← $ReLU = max(0, z)$

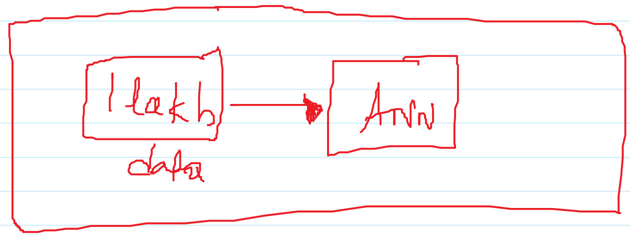
$max(0, 2.5) \rightarrow 2.5$

o/p layer neuron? ✗

o/p — sigmoid, softmax





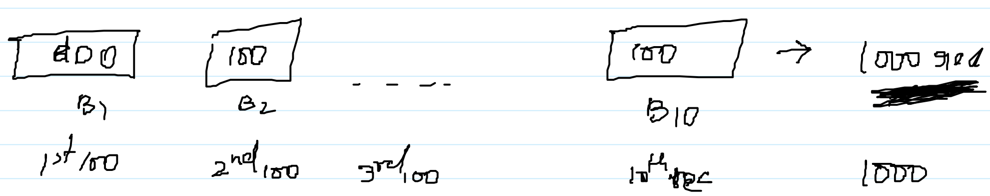


(1 lakh, 3 col)

CPU ↓ speed ↓ hang ↓ long

1000 rec → Train

1000 rec ✓ Epoch,



V

B1 B2 B10

1st/100 2nd/100 3rd/100 10th rec 1000

10

10 rec

Batch size = 5

Epoch 1 $\boxed{R_1 R_2 R_3 R_4 R_5}$ B1 $\boxed{R_6 R_7 R_8 R_9 R_{10}}$ B2 → All Records are trained only time

until we get mini error

Epoch 2 $\boxed{R_3 R_5 R_7 R_9 R_1}$ $\boxed{R_2 R_4 R_6 R_8 R_{10}}$ → Trained only time

⋮

Epoch n